Paul Leger: Publications (*ORCID:* 0000-0003-0969-5139)

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NOTE: For conferences/workshops, I use the CORE ranking, which belongs **CO**mputing **RE**search and Education Association of Austral-asia (near to the publication year). CORE is available on http://portal.core.edu.au/conf-ranks.

Journals (Indexed)

Canquil, Diego, Daniel SanMartín, Irene Inoquio-Renteria, and Paul Leger (Jan. 2026). "A Systematic Literature Review on Machine Learning Techniques for Predicting Household Water Consumption". In: *Water Resources Management* 40.2, p. 83. DOI: 10.1007/s11269-026-04496-4. **JCR-WOS**.

Leger, Paul, David Moreno-Lumbreras, Felipe Ruiz, Nicolás Sepúlveda, Ismael Figueroa, and Nicolás Cardozo (Feb. 2026). "The Stackage Repository: An Exploratory Study of its Evolution". In: *Journal of Computer Languages* 86, p. 101386. DOI: 10.1016/j.cola.2026.101386. **JCR-WOS**.

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Abdi, Farshid, Shaghayegh Abolmakarem, Amir Karbassi-Yazdi, Paul Leger, Yong Tan, and Giuliani Coluccio (Jan. 2025). "Predicting Patients' Revisit Intention Based on Satisfaction Scores: Combination of Penalized Regression and Neural Networks". In: *IEEE ACCESS* 13, pp. 2783–2800. DOI: 10.1109/ACCESS.2024.3522767. **JCR-WOS**.

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Leger, Paul, Grecia Gálvez, Lino Cubillos, Diego Cosmelli, Milton Inostroza, Éric Tanter, Gina Luci, and Jorge Soto Andrade (Mar. 2014). "ECOCAM, un sistema computacional adaptable al contexto para promover estrategias de cálculo mental: características de su diseño y resultados preliminares". In: *Revista Latinoamericana de Investigación en Matemática Educativa* 17.1, pp. 33–58. DOI: 10.12802/relime.13.1712. **JCR-WOS**.

Ramírez, Pablo, Paul Leger, and Andres Vallone (July 2014). "Un Modelo Flexible para la Simulación de Distribución de Ciudades". In: *Revista chilena de ingeniería* 22.3, pp. 363–373. DOI: 10.4067/S0718-33052014000300007. **SCOPUS**.

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Ogino, Toshihiro, Hiroaki Fukuda, and Paul Leger (July 2025). "Stepwise Distributed Execution for Load Reduction in Machine Learning Systems". In: *International Symposium on Parallel and Distributed Computing (ISPDC)*. Rennes, France. **To appear. SCOPUS. Rank C (CORE ranking)**.

You, Koshi, Hiroaki Fukuda, and Paul Leger (Apr. 2025). "The Impact of Context-Oriented Programming on Declarative UI Design in React". In: *Evaluation of Novel Approaches to Software Engineering (ENASE)*. Porto, Portugal, pp. 700–707. DOI: 10.5220/0013432300003928. **SCOPUS. Rank B (CORE ranking)**.

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Liu, Shijin, Hiroaki Fukuda, and Paul Leger (Nov. 2023a). "A RF-based Low Rate DDoS Attack Real-time Detection System". In: *33rd International Telecommunication Networks and Applications Conference (ITNAC)*. Melbourne, Australia: IEEE, pp. 304–309. DOI: 10.1109/ITNAC59571.2023.10368543. **SCOPUS. Rank C-Australasian (CORE ranking)**.

— (Mar. 2023b). "Real-time DDoS Attack Defense System in SDN Using LSSOM". In: *26th Conference on Innovation in Clouds, Internet and Networks and Workshops (ICIN)*. Paris, France, pp. 69–73. DOI: 10.1109/ICIN56760.2023.10073509. **SCOPUS. Rank National-France (CORE ranking)**.

Leger, Paul, Felipe Ruiz, Nicolás Sepúlveda, and Ismael Figueroa (Apr. 2022). "Evolution of a Haskell Repository and its Use of Monads: An Exploratory Study of Stackage". In: *Proceedings of the 37th Annual ACM Symposium on Applied Computing (SAC 2022)*. Virtual Conference: ACM Press, pp. 1475–1482. DOI: 10.1145/3477314.3506982. **SCOPUS. Rank B (CORE ranking)**.

Fukuda, Hiroaki, Ryota Gunji, Tadahiro Hasegawa, Paul Leger, and Ismael Figueroa (Jan. 2020). "Toward distributed Streaming data Sharing Manager for Autonomous Robot Control". In: *IEEE/SICE International Symposium on System Integration (SII)*. Honolulu, USA: IEEE, pp. 862–866. DOI: 10.1109/SII46433.2020.9025920. **SCOPUS. Superseded by Sensors**.

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Jaque, Dayana, Paul Leger, Felipe Machorro, and Vanessa Romero Ortiz (Nov. 2020). "Indicadores de Capital Intelectual para Gestionar el Desempeño en Universidades Chilenas". In: *36th IBIMA International Conference*. Granada, España: IBIMA.

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Machorro, Felipe, Vanessa Romero Ortiz, Paul Leger, and Dayana Jaque (Nov. 2020). "Efecto Mediador de la Gestión del Conocimiento entre las Tecnologías de Información y el Desempeño Organizacional de las Instituciones de Educación Superior". In: *36th IBIMA International Conference*. Granada, España: IBIMA.

Manzano, Carlos, Claudio Meneses, and Paul Leger (Nov. 2020). "Empirical Comparison of Supervised Algorithms for Ransomware Identification on Network Traffic". In: *2020 - The International Conference of the Chilean Computer Science Society (SCCC)*. Coquimbo, Chile: IEEE, pp. 1–7. DOI: 10.1109/SCCC51225.2020.9281283. **SCOPUS**.

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Figueroa, Ismael, Bruno García, and Paul Leger (Sept. 2018). "Towards Progressive Verification in Dafny". In: *Proceedings of the XXII Brazilian Symposium on Programming Languages (SBLP 2018)*. São Carlos, Brazil: ACM Press, pp. 90–97. DOI: 10.1145/3264637.3264649. **SCOPUS**.

Leger, Paul and Hiroaki Fukuda (Apr. 2017). "Sync/CC: Continuations and Aspects to Tame Callback Dependencies on JavaScript Handlers". In: *Proceedings of the 32nd Annual ACM Symposium on Applied Computing (SAC 2017)*. Marrakech, Morocco: ACM Press, pp. 1245–1250. DOI: 10.1145/3019612.3019783. **Rank B (CORE ranking). Superseded by JUCS (Journal of Universal Computer Science)**.

Sabin, Mihaela, Barbara Viola, John Impagliazzo, Renzo Angles, Mariela Curiel, Paul Leger, Jorge Murillo, Hernán Nina, José Antonio Pow-Sang, and Ignacio Trejos (July 2016). "Latin American Perspectives to Internationalize Undergraduate Information Technology Education". In: *Proceedings of the 2016 ACM Conference on Innovation and Technology in Computer Science Education (ITiCSE 2016)*. Arequipa, Peru: ACM Press, pp. 1–22. DOI: 10.1145/3024906.3029847. **Rank A (CORE ranking). NOTE: Selected in "the best of ourselves" Chilean JCC-2019**.

Rocha, Wiliam, Hiroaki Fukuda, and Paul Leger (Dec. 2015). "Modular Asynchronous Web Programming: Advantages and Challenges". In: *Proceedings of 9th International Conference on Bio-inspired Information and Communications Technologies (BICT 2015)*. ACM Press. DOI: 10.4108/eai.3-12-2015.2262472. **SCOPUS**.

Vidal, Cristian, David Benavides, Paul Leger, José Galindo, and Hiroki Fukuda (Dec. 2015). "Mixing of Join Point Interfaces and Feature-Oriented Programming for Modular Software Product Line". In: *Proceedings of 9th International Conference on Bio-inspired Information and Communications Technologies (BICT 2015)*. New York, USA: ACM Press. DOI: 10.4108/eai.3-12-2015.2262534. **A mirror version (with the same DOI) was published in the journal EAI Endorsed Transactions on Scalable Information Systems**.

Leger, Paul and Éric Tanter (Sept. 2010). "An Open Trace-Based Mechanism". In: *Proceedings of the 14th Brazilian Symposium on Programming Languages (SBLP 2010)*. Ed. by Jonathan Aldrich and Ricardo Massa. Salvador - Bahia, Brazil, pp. 123–138.

Toledo, Rodolfo, Paul Leger, and Éric Tanter (Mar. 2010). "AspectScript: Expressive Aspects for the Web". In: *Proceedings of the 9th ACM International Conference on Aspect-Oriented Software Development (AOSD 2010)*. Rennes and Saint Malo, France: ACM Press, pp. 13–24. DOI: 10.1145/1739230.1739233. **Rank A (CORE ranking)**.

Workshops (SCOPUS)

Fukuda, Hiroaki, Paul Leger, and Nicolás Cardozo (June 2022). "Layer Activation Mechanism for Asynchronous Executions in JavaScript". In: *Proceedings of the 14th International Workshop on Context-Oriented Programming and Advanced Modularity (COP 22)*. Berlin, Germany: ACM Press, pp. 1–8. DOI: 10.1145/3570353.3570354. **SCOPUS**.

Cleveland, Margareth and Paul Leger (Nov. 2020). "A Collaborative Learning Strategy in an MIS Development Course Using Case Method in Engineering in Information and Management Control". In: *Congreso Chileno de TICs para la Educación (TICXED)*. Coquimbo, Chile: IEEE, pp. 1–5. DOI: 10.1109/SCCC51225.2020.9281192. **SCOPUS**.

Leger, Paul, Hidehiko Masuhara, and Ismael Figueroa (July 2020). "Interfaces for Modular Reasoning in Context-Oriented Programming". In: *Proceedings of the 12th International Workshop on Context-Oriented Programming and Advanced Modularity (COP 20)*. 3. Virtual Event, USA: ACM Press, pp. 1–7. DOI: 10.1145/3422584.3423152. **SCOPUS**.

Leger, Paul and Hiroaki Fukuda (Mar. 2016). "Using Continuations and Aspects to Tame Asynchronous Programming on the Web". In: *Proceedings of the Workshop on Foundations of Aspect-Oriented Languages (FOAL 2016)*. Malaga, Spain: ACM Press, pp. 79–82. DOI: 10.1145/2892664.2892675. **Rank C (CORE ranking). Superseded by JUCS (Journal of Universal Computer Science)**.

Fukuda, Hiroaki and Paul Leger (June 2015c). "Proposals for Modular Asynchronous Web Programming: Issues and Challenges". In: *Proceedings on Pervasive Web Technologies, Trends and Challenges (PEWET*

2015). Vol. 9396. Lecture Notes in Computer Science. Rotterdam, the Netherlands: Springer, pp. 91–102. DOI: 10.1007/978-3-319-24800-4_8. **Published on Lectures Notes in Computer Science**.

Silva, Cristian Vidal, David Benavides, José Ángel Galindo, Paul Leger, Rodolfo Villarroel, and Sebastián Valenzuela (Nov. 2015). "JPI Feature Models - Exploring a JPI and FOP Symbiosis". In: *WASE 2015 Workshop on Advanced Software Engineering*. Santiago, Chile: IEEE, pp. 1–6. DOI: 10.1109/SCCC.2015.7416583.

Leger, Paul and Éric Tanter (Mar. 2012). "A Self-Replication Algorithm to Flexibly Match Execution Traces". In: *Proceedings of the 11th Workshop on Foundations of Aspect-Oriented Languages (FOAL 2012)*. Potsdam, Germany: ACM Press, pp. 27–32. DOI: 10.1145/2162010.2162019. **Rank C (CORE ranking)**.

Congresses (*Conferences/Workshops not indexed*)

Note: These papers are just presentations of technical reports or working papers.

Moreno-Lumbreras, David, Paul Leger, Sergio Montes-León, Jesus Gonzalez-Barahona, and Gregorio Robles (Sept. 2025). "Evaluating the Comprehension of the Stackage Ecosystem: A Comparison between VR and 2D Visualizations". In: *Registered Report: 2025 IEEE International Conference on Software Maintenance and Evolution (ICSME)*. Auckland, New Zealand. DOI: 10.2139/ssrn.5362665. **Registered Report**.

Espinoza, Francis and Paul Leger (Dec. 2024). "Using Artificial Intelligence to Democratise the Information". In: *International Conference Promoting Democracy and the Rule of Law – Global Perspectives Santiago de Chile*. Santiago, Chile.

Ortiz, Mario, Paul Leger, and Carmen Hidalgo-Alcázar (Sept. 2022). "Clasificación de Usuarios en Perfiles de Compradores en el Sector Inmobiliario: Análisis Mediante una Experiencia Virtual". In: *XXXIII Congreso de Marketing (AEMARK 2022)*. Valencia, Spain.

López, Manuela, Oswaldo Terán, and Paul Leger (Sept. 2021). "Modeling Chinese Cross-Border B2C E-commerce in Spain: A Multi-Agent Based Simulation Approach". In: *XXXII Congreso de Marketing (AEMARK 2021)*. Baeza y Úbeda, Spain.

Figueroa, Ismael, Paulina Guitierrez, Natalia Araya, and Paul Leger (Nov. 2020). "Estudio Indagatorio sobre la Usabilidad Percibida de los Sistemas de Información Contable y Financieros". In: *Conferencia Académica Permanente de Investigación Contable (CAPIC)*. La Serena, Chile.

Pinto, Domingo, Diego Cortes, Paul Leger, and Manuela López (Sept. 2020). "Una Arquitectura de una Implementación Abierta de un Modelo Basado en Agentes para Campañas de WOM Marketing". In: *XI Congreso Internacional de Computación e Informática del Norte de Chile (Infonor 2020)*. Atacama, Chile.

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